

L04 REFLECTIVE JOURNAL

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ITAI 1371 — Module 04: Exploratory Data Analysis

Lab 4 - Reflection

This lab helped better understand the purpose of Exploratory Data Analysis (EDA) and why it is important before building a machine learning model. I learned that EDA is like looking for patterns and clues in a dataset before trying to make predictions. Using the Titanic dataset, I was able to see how statistics and visualizations can reveal information about the passengers. The graphs showed basic information like the average age, survival rate, and range of ticket fares. but, the visualizations made some of the relationships much easier to know.

One of the biggest things I noticed was how strongly survival was related to passenger class and gender. First-class passengers had a higher survival rate than third-class passengers, while females survived at a much higher rate than males. The age visualization also showed a noticeable number of children among the survivors. I also explored survival based on the port where passengers boarded and compared ticket fares between passengers who survived and those who did not. These experiments showed me that different variables can have noticeable relationships with the outcome we are trying to understand.

I also learned why visualizations are useful alongside statistics. A mean or count can give useful information, but it does not always show how the data is distributed or whether there are unusual values. Graphs can make differences, patterns, and outliers much easier to notice. But also it can me things ewasier to understand for someone who may not know exactly what they are looking at. Overall, this lab showed me that EDA is an important part of working with data because it helps us understand what the dataset is telling us before we use it to build a machine learning model. It also showed me that looking at the data from multiple perspectives can reveal information that may not be obvious from numbers alone.

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Introduction

Module 04 introduced Exploratory Data Analysis (EDA) and its role in understanding data before modeling. Using the Titanic dataset in Google Colab, I used Python, Pandas, Matplotlib, and Seaborn to examine the data, calculate statistics, create visualizations, and explore additional relationships. I learned that EDA is more than simply viewing data; it helps identify patterns, data-quality issues, and questions that should be investigated before modeling. Understanding EDA, I examined measures such as mean, standard deviation, minimum, maximum, and quartiles. I also learned that statistics alone may not reveal the full story. Visualizations of survival, passenger class, gender, and age made relationships and differences easier to recognize. The lab also reinforced the importance of checking for missing values and understanding the structure of a dataset before analysis.

Experimentation and Learning

I completed two additional experiments: survival by port of embarkation and fare distribution by survival. The fare boxplot suggested an association between higher fares and survival, but I learned that an observed relationship does not prove causation. These experiments helped me understand that EDA involves asking questions and using visualizations to investigate them, not simply producing charts.

Challenges and Problem-Solving

I encountered challenges with Python code structure and notebook execution in Google Colab. I learned to examine errors carefully, correct the code, and rerun the appropriate cells.

This reinforced the importance of executing notebook cells systematically and reviewing outputs. I also learned that successfully running code is not enough; I must understand and interpret what the results mean.

Connection to Previous Learning

L04 connected with my previous machine-learning work because data exploration is an important step before model development. It also relates to my accounting, banking, and finance background, where analyzing patterns, identifying unusual transactions, and investigating relationships are important. EDA can similarly help identify trends, anomalies, and areas requiring further investigation.

How My Thinking Has Changed

Previously, I focused more on numerical results. This lab helped me understand that visualization and interpretation are equally important. A statistic may summarize the data, while a visualization can reveal distributions, differences, and patterns that numbers alone may not show.

My understanding of the process is now:

Question → Data → Exploration → Visualization → Interpretation → Further Questions

Conclusion

The most important lesson from Module 04 is that understanding the data should come before drawing conclusions or building models. The lab improved my ability to use statistics, visualization, experimentation, and critical thinking together. I will apply this systematic approach to future data-science and financial-analysis projects.

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Reflection

This Module 04 Lab gave me a better understanding of Exploratory Data Analysis (EDA). When I studied Machine Learning before, I mostly focused on whether the code worked and what results the model gave. This lab made me realize that understanding the dataset before building a model is also very important.

Using the Titanic dataset, I learned that statistics and visualization can help us understand data in different ways. For example, the `describe()` method gives information such as average age, survival rate, and fare range. However, these numbers do not show all the patterns in the data. Visualizations such as "Survival by Gender" and "Fare by Survival" made the differences between groups easier to see. I noticed that female passengers had higher survival rates and that survivors generally paid higher fares.

The lab also made me pay more attention to missing values. The Titanic dataset had missing values in "Age" and "Cabin." In the past, I might have only checked if the code had errors. Now, I pay more attention to problems in the data itself. If the dataset has problems, it can also affect the Machine Learning model later.

I think this lab connects well with the Machine Learning workflow I learned before. Machine Learning is not only about choosing a model and checking its accuracy. We also need to understand and check the data before modeling. EDA helps us find patterns, missing values, and possible relationships between variables.

My biggest takeaway from this lab is that my view of EDA has changed. I no longer see it as just making a few charts. I now see it to understand the dataset before building a model. In

future Machine Learning projects, I will spend more time understanding the data before moving on to the modeling stage.

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EDA Lab: Reflection

Keith Jones

This lab was built around a simple premise — take one dataset and ask it several different questions, one plot at a time — but the thing I noticed by the end was how consistently the same shape of question kept coming back, no matter which variable I was looking at.

Every experiment followed the same arc. First, plot the raw relationship: a countplot of survival by port, a histogram of age split by outcome, a boxplot of fare against survival. Each one, on its own, showed something. Cherbourg passengers looked like they survived at a higher rate. Young children showed a visible spike in survival. Survivors had a noticeably higher median fare. Taken individually, each of these felt like a discovery.

But the more plots I made, the more I noticed the same name showing up as a plausible explanation underneath all of them: passenger class. Fare is essentially a proxy for class. Port of embarkation correlates with class, because different ports fed different proportions of first-, second-, and third-class passengers onto the ship. Even the "women and children first" pattern likely interacted with class, since lifeboat access wasn't distributed evenly across decks. So, a plot that looks like it's telling me "Fare mattered" or "port mattered" might really be telling me "Class mattered" wearing a different variable's name.

That was the most useful thing I took from this lab, and it wasn't something any single plot could have taught me by itself. It only became visible by comparing plots to each other and noticing they kept pointing at the same underlying factor. A countplot answers "is there a relationship here?" It cannot answer "is this relationship real, or is it riding on the back of something else?" That second question is the one I now think EDA is actually for. The first pass at any plot is descriptive. The real work starts when you ask why the description looks the way it does.

I also noticed how much interpretation depends on how a plot is built, not just what data goes into it. The FacetGrid histogram of age looked meaningfully different depending on whether the two panels were compared by raw count or by proportion, since more passengers died than survived overall. The fare boxplot was nearly unreadable until I capped or log-scaled the axis, because a handful of very expensive tickets flattened everything else near zero. None of the underlying data changed — only the lens did — and the lens changed what conclusion felt obvious. That's a slightly uncomfortable thing to sit with: the same numbers can visually argue two different points depending on formatting choices, which means part of doing EDA honestly is being suspicious of your own first impression of a chart.

By the final markdown question — describing "the kind of person" most likely to survive — I could answer it with actual confidence, not because one plot told me, but because three separate plots (port, age, fare) all converged on the same underlying story: young, female, and in first class. No individual chart proved that. The pattern across all of them did.